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Nakayama Algebras

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1 Introduction

One of our main goals in the course was to find a characterisation for all indecomposables. Nakayama algebras is a class of algebras where we have a neat characterisation for the same. In this report, we will build up to understand what a Nakayama Algebra is, looking at the composition series, radical series and socle series of a module, uniserial modules, left and right algebras, and ordinary quivers, and then going on to understand indecomposables and almost split sequences in Nakayama Algebras.

2 Composition Series and Composition Length

Definition 1. Given a finite dimensional K -algebra, A , and M in $\text{mod-}A$, the chain

$$0 = M_0 \subset M_1 \subset M_2 \subset \dots \subset M_m = M$$

of submodules of M such that M_{j+1}/M_j is simple for $j = 0, 1, \dots, m-1$ is called a **composition series** of M .

Theorem 2. Jordan - Hölder Theorem

Given two composition series for a module M ,

$$0 = M_0 \subset M_1 \subset M_2 \subset \dots \subset M_m = M \text{ and } 0 = N_0 \subset N_1 \subset N_2 \subset \dots \subset N_n = M$$

we have $m = n$ and there is a permutation σ of $\{1, \dots, m\}$ such that $M_{j+1}/M_j \cong N_{\sigma(j+1)}/N_{\sigma(j)}$ for any $j = 0, 1, \dots, m-1$.

Hence, the number of modules in the composition series depends only on M . We call this the **composition length** of M , and denote it by $l(M)$.

3 Radical Series, Socle Series and Loewy Length

Definition 3. For a right A -module, M , the (Jacobson) **radical** is the intersection of all the maximal submodules of M , and is denoted by $\text{rad}M$.

Definition 4. For a right A -module, M the **socle** is the submodule of M that is generated by all simple submodules of M , and is denoted by $\text{soc}M$.

Definition 5. For a right A -module, M , $\text{top}M = M/\text{rad}M$, is called the **top** of M .

3.1 Radical Series or Descending Loewy Series of M

Definition 6. For an A -module, M , the decreasing sequence of submodules of M ,

$$M \supset \text{rad}M \supset \text{rad}^2M \supset \dots \supset \text{rad}^iM \supset \dots \supset 0$$

is called the **radical series** or **descending Loewy series** of M .

Since M is finite dimensional, the series is finite. Let m be the least positive integer such that $\text{rad}^mM = 0$. This m is called the **length of the radical series**, and is denoted by $rl(M)$.

3.2 Socle Series or Ascending Loewy Series of M

For an integer $i \geq 0$, we define $\text{soc}^i(M)$ inductively, with $\text{soc}^0 M = 0$. Let p be the quotient map, $p : M \rightarrow M/\text{soc}^i M$. Then $\text{soc}^{i+1} M = p^{-1}(\text{soc}(M/\text{soc}^i M))$. Note that $\text{soc}^{i+1} M \supset \text{soc}^i M$.

Definition 7. For an A -module, M , the increasing sequence of submodules of M ,

$$0 = \text{soc}^0 M \subset \text{soc} M = \text{soc}^1 M \subset \text{soc}^2 M \subset \dots \subset \text{soc}^j M \subset \dots \subset M$$

is called the **socle series** or **ascending Loewy series** of M .

Since M has finite composition length, the series is finite. Let m be the least positive integer such that $\text{soc}^m M = M$. This m is called the **length of the socle series**, and is denoted by $sl(M)$.

3.3 Relation between $rl(M)$ and $sl(M)$ and Loewy Length

In general, the radical series and the socle series don't coincide. However, their lengths coincide! The proof for $rl(M) = sl(M)$ proceeds via a fibered product, using properties of duals and induction. We call this common value of $rl(M)$ and $sl(M)$ as the **Loewy length** of M , and denote it by $ll(M)$.

[Alfred Loewy \(1873 - 1935\)](#)



Alfred Loewy was a German mathematician, after whom Loewy rings, Loewy length, Loewy decomposition and the Loewy series are named. He worked with linear groups, algebraic differential equations, and actuarial mathematics. In 1935, he was forced into retirement as he was a Jew. Wolfgang Krull, one of the mathematicians after whom the Krull-Schmidt theorem is named, was Loewy's student.

3.4 Properties of rad

The following are some important properties of rad , which are used in later proofs:

1. $M\text{rad}A = \text{rad}M$
2. $\text{rad}(M \oplus N) = \text{rad}M \oplus \text{rad}N$
3. Given an A -module epimorphism $f : M_A \rightarrow N_A$, $f(\text{rad}^i M) = \text{rad}^i N$ for every $i \geq 0$.

Proof. $f(\text{rad}M) = f(M\text{rad}A) = f(M)\text{rad}A = N\text{rad}A = \text{rad}N$ □

4. Given a decomposition of a module $M = M_1 \oplus \dots \oplus M_m$, we have

$$ll(M) = \max\{ll(M_1), \dots, ll(M_m)\}$$

Proof. follows from (2) □

4 Uniserial Modules

Definition 8. An A -module M_A is called **uniserial** if it has a unique composition series.

Some important properties of uniserial modules are:

- If M is uniserial, every submodule of M is also uniserial.
- If M is uniserial, every quotient of M is also uniserial.
- Uniserial modules are determined up to isomorphism by their composition series.
- The dual of a uniserial right module is a uniserial left module.
- Uniserial modules have simple tops, and simple socles.
- All uniserial modules are indecomposable.

Lemma 9. The following are equivalent for M , a right A -module:

1. M is uniserial.
2. The radical series, $M \supset \text{rad}M \supset \dots \supset 0$ is a composition series.
3. The socle series, $0 \subset \text{soc}M \subset \dots \subset M$ is a composition series.
4. $l(m) = ll(m)$

Proof. We first show $(1 \iff 2)$. Then we can get $(1 \iff 3)$ analogously.

$(1 \implies 2)$

Induction on composition length $l(M)$ of M :

- Base Case: $l(M) = 1 \implies M$ is simple. In this case the statement is trivial.
- Induction Step: Consider a uniserial module M , with $l(M) = t$. Since M is uniserial, it has a unique maximal submodule, coinciding with $\text{rad}M$, which is also uniserial. The result now follows from the induction hypothesis.

$(2 \implies 1)$

Assume $M = M_0 \supset M_1 \dots \supset M_t = 0$ and $M = N_0 \supset N_1 \supset \dots \supset N_t = 0$ are two composition series for M . Then, $M_i = N_i = \text{rad}^i M$ for every $0 \leq i \leq t$, by induction on i .

- Base Case: The result holds trivially for $i = 0$, as $\text{rad}^0 M = M = M_0 = N_0$.
- Induction Step: $\text{rad}^i M / \text{rad}^{i+1} M$ is simple $\implies \text{rad}^i M$ has a unique maximal submodule, which is necessarily $\text{rad}^{i+1} M \implies M_{i+1} = N_{i+1} = \text{rad}^{i+1} M$.

$(4 \implies 2)$

- If N is an A -submodule of M , then $l(M) = l(N) + l(M/N)$.
- Hence, $m = l(M) = ll(M) = \sum_{i=0}^{m-1} l(\text{rad}^i M / \text{rad}^{i+1} M)$
 $\implies l(\text{rad}^i M / \text{rad}^{i+1} M) = 1$ for $i = 0, 1, \dots, m-1$ (since they can't be 0).
- This implies that each $\text{rad}^i M / \text{rad}^{i+1} M$ is simple, implying that the radical series is a composition series.

$(2 \implies 4)$ is obvious. It follows from the definitions.

□

5 Right Serial and Left Serial Algebras

Definition 10. An algebra A is **right serial** if every indecomposable projective right A -module is uniserial.

Definition 11. An algebra A is **left serial** if every indecomposable projective left A -module is uniserial.

Equivalently,

- A is right serial if every indecomposable injective left A -module is uniserial.
- A is left serial if every indecomposable injective right A -module is uniserial.

Now let us look at a characterisation of right serial algebras.

Lemma 12. An algebra A , is right serial if and only if for every indecomposable projective right module P , the module $\text{rad}P/\text{rad}^2P$ is simple or zero.

Idea of Proof.

($\Rightarrow$)

$$M \supset \text{rad}P \supset \text{rad}^2P \supset \dots \supset \text{rad}^iP \supset \dots \supset 0$$

The radical series of uniserial modules is a composition series. Hence $\text{rad}P/\text{rad}^2P$ must be simple or zero. This implication is straightforward.

($\Leftarrow$)

We can show that the radical series is a composition series by proving that $\text{rad}^{i-1}P/\text{rad}^iP$ is simple or zero for $i \geq 1$, using induction on i . The idea of the induction step is as follows:

Let $f : P' \rightarrow \text{rad}^{i-1}P$ be a projective cover, and $p : \text{rad}^{i-1}P \rightarrow \text{rad}^{i-1}P/\text{rad}^iP$. $pf : P' \rightarrow \text{rad}^{i-1}P/\text{rad}^iP$ is then a projective cover. Since $\text{rad}^{i-1}P/\text{rad}^iP$ is simple, P' is indecomposable. f restricts to f_1 and f_2 , giving rise to $\bar{f}$ on passing to cokernels.

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{rad}^2P' & \longrightarrow & \text{rad}P' & \longrightarrow & \text{rad}P'/\text{rad}^2P' \longrightarrow 0 \\ & & \downarrow f_2 & & \downarrow f_1 & & \downarrow \bar{f} \\ 0 & \longrightarrow & \text{rad}^{i+1}P & \longrightarrow & \text{rad}^iP & \longrightarrow & \text{rad}^iP/\text{rad}^{i+1}P \longrightarrow 0 \end{array}$$

P' is indecomposable and by the hypothesis for any such P' , $\text{rad}P'/\text{rad}^2P'$ is simple or zero. Hence, via $\bar{f}$ we get that $\text{rad}^iP/\text{rad}^{i+1}P$ is also simple or zero.

6 Ordinary Quiver of an Algebra

Recall that, an algebra is called **basic**, if $e_iA \not\cong e_jA$ for all $i \neq j$. Also, an algebra is called **connected** if the only central idempotents of the algebra are 0 and 1.

Definition 13. Let A be a basic and connected finite dimensional K -algebra and $\{e_1, e_2, \dots, e_n\}$ be a complete set of primitive orthogonal idempotents of A . The **ordinary quiver** of A , Q_A is defined as follows:

1. The n vertices of the quiver correspond to $e_1, e_2, \dots, e_n$.
2. Given $a, b \in (Q_A)_0$, the arrows $\alpha : a \rightarrow b$ correspond to the vectors in a basis of $e_a(\text{rad}A/\text{rad}^2A)e_b$. The dimension of this vector space is the number of arrows going from a to b .

- If A is a basic and connected finite-dimensional algebra, then Q_A is connected.

For example,

$$\text{Consider the algebra } A = \begin{bmatrix} K & 0 & 0 \\ K & K & 0 \\ K & 0 & K \end{bmatrix}$$

of lower triangular matrices $\lambda_{ij} \in M_3(K)$, with $\lambda_{32} = 0$ and $\lambda_{pq} = 0$ for $p > q$.

A complete set of primitive orthogonal idempotents is:

$$e_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, e_3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \text{ Also, } \text{rad}A = \begin{bmatrix} 0 & 0 & 0 \\ K & 0 & 0 \\ K & 0 & 0 \end{bmatrix}$$

and $\text{rad}^2 A = 0$. $e_2(\text{rad}A)e_1$ and $e_3(\text{rad}A)e_1$ are one-dimensional, and all other spaces of the form $e_i(\text{rad}A)e_j$ are 0, giving Q_A as,

$$\begin{array}{ccc} & 1 & \\ \alpha \nearrow & & \nwarrow \beta \\ 2 & & 3 \end{array}$$

Observe that the projectives and injectives are as follows:

$$P(1) = \begin{bmatrix} K & 0 & 0 \end{bmatrix}, P(2) = \begin{bmatrix} K & K & 0 \end{bmatrix}, P(3) = \begin{bmatrix} K & 0 & K \end{bmatrix}$$

$$I(1) = \begin{bmatrix} K & K & K \end{bmatrix}, I(2) = \begin{bmatrix} 0 & K & 0 \end{bmatrix}, I(3) = \begin{bmatrix} 0 & 0 & K \end{bmatrix}$$

On looking at the composition series of these, we observe that all projectives have unique composition series, while on looking at injectives, we see that $I(1)$ doesn't. Hence, this is an example of an algebra that is not both left serial and right serial.

6.1 Characterization of Basic Right Serial Algebras

Theorem 14. *A basic K -algebra is right serial if and only if, for every point a of its ordinary quiver Q_A , there exists at most one arrow of source a .*

Proof. (Follows from Lemma 12)

- $\text{rad}P(a)/\text{rad}^2P(a) = e_a(\text{rad}A/\text{rad}^2A)$ is simple or zero, and hence as a K -vector space has a dimension of 0 or 1.
- By definition of Q_A , there is at most one $b \in (Q_A)_0$ such that $e_a(\text{rad}A/\text{rad}^2A)e_b \neq 0$, i.e. there is at most one such b with an arrow $a \rightarrow b$.

□

Tadashi Nakayama (1912 - 1964)

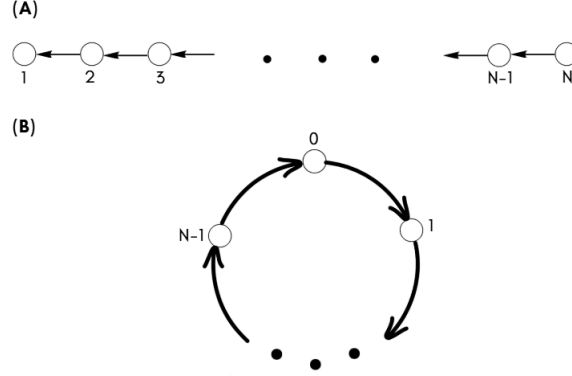


Tadashi Nakayama was a Japanese Mathematician who made many important contributions to Representation Theory. Nakayama's lemma, Nakayama algebras, Nakayama's conjectures and the Murnaghan-Nakayama rule are named after him. He also worked on Frobenius rings and Galois Theory. In his final days, he was learning Grothendieck's approach to Algebraic Geometry. While he spent most of his life in Japan, he was in Princeton for 2 years, where he met Brauer and Artin.

7 Nakayama Algebras

Definition 15. An algebra, A , is called a **Nakayama Algebra** if it is both right and left serial.

Theorem 16. A basic and connected algebra A is a Nakayama Algebra if and only if its ordinary quiver Q_A is one of the following:



The proof follows from the previous characterization.

7.1 Indecomposables in Nakayama Algebras

Lemma 17. Let A be an algebra, and J , a proper ideal of A .

1. If A is right serial, then so is A/J .
2. If A is a Nakayama Algebra, then so is A/J .

Proof. If $A_A = \bigoplus_{i=1}^n P_i$ is a direct sum decomposition of indecomposables, we have $A/J = \bigoplus_{i=1}^n P_i/P_i J$ as the direct sum decomposition of A/J , with each $P_i/P_i J$ either an indecomposable or zero, implying that A/J is also right serial.

(2) follows as a consequence of (1) and it's dual. $\square$

Theorem 18. Let A be a basic and connected Nakayama algebra. For any indecomposable A -module M , there is an indecomposable projective P and an integer t with $1 \leq t \leq \text{Il}(P)$ such that $M \cong P/\text{rad}^t P$.

Proof. Each $P/\text{rad}^t P$ is uniserial and hence indecomposable. Take $t = \text{Il}(M)$. This gives $\text{rad}^t M = 0 \implies M \text{rad}^t A = 0$. Hence, M has the structure of an $A/\text{rad}^t A$ module, which is itself a Nakayama algebra. We will now work in this module, which has the decomposition,

$$A/\text{rad}^t A \cong \bigoplus_{i=1}^n (P_i/P_i \text{rad}^t A) = \bigoplus_{i=1}^n (P_i/\text{rad}^t P_i)$$

Let $f : \bigoplus_{j=1}^r P'_j \rightarrow M$ be a projective cover of M in $A/\text{rad}^t A$, with P'_j indecomposable for all $j = 1, 2, \dots, r$. Then, $t = \text{Il}(A/\text{rad}^t A) = \max\{\text{Il}(P'_1), \dots, \text{Il}(P'_r)\} \geq \text{Il}(M) = t$. This implies that $\exists j \in \{1, 2, \dots, r\}$ such that $\text{Il}(P'_j) = t$. Let f_j be the function we get on restricting f to P'_j . If no such j gives rise to a monomorphism, f_j , then $\text{Il}(\text{Im } f_j) < t$ for all $j = 1, 2, \dots, r$, which implies $\text{Il}(M) < t$, which is not possible. Hence we have a $q \in \{1, 2, \dots, r\}$ such that $f_q : P'_q \rightarrow M$ is a monomorphism. This f_q is an isomorphism, which gives us $M \cong P'_q \cong P_i/\text{rad}^t P_i$ for some $1 \leq i \leq n$, concluding the proof. $\square$

★ As a consequence, **all indecomposables are uniserial!** We previously saw that all uniserial modules are indecomposable. Hence, this gives us a characterisation for all indecomposables in Nakayama Algebras!

7.2 Almost Split Sequences for Nakayama Algebras

Theorem 19. *For an indecomposable nonprojective A -module, $M \cong P/\text{rad}^t P$, the following is an almost split sequence.*

$$0 \rightarrow \text{rad}P/\text{rad}^{t+1}P \xrightarrow{[q, i]^t} (\text{rad}P/\text{rad}^t P) \oplus (P/\text{rad}^{t+1}P) \xrightarrow{[-j, p]} P/\text{rad}^t P \rightarrow 0$$

where q, p are the canonical epimorphisms (quotient maps), and i, j are inclusion homomorphisms.

Idea of Proof.

- Let $v : V \rightarrow M$ be a non isomorphism where V is an indecomposable module.
- In each case of v (either surjective or not), we can get a map that shows that v must factor through $g = [-j, p]$, proving that g is right almost split.
- M being indecomposable implies that $f = [q, i]^t$ is left almost split.

Corollary 20. *For every indecomposable nonprojective A -module M , we have $l(\tau M) = l(M)$.*

★ All almost split sequences in the module category of Nakayama Algebras have at most two indecomposable middle terms (follows from Theorem 19)

7.3 Where are Nakayama Algebras used?

- They are used to study regular modules over representation-infinite hereditary algebras.
- Derived categories of Nakayama Algebras are used in **algebraic geometry**.
- Linear Nakayama Algebras correspond to the **combinatorics** of Dyck paths via their AR Quiver.

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